



Machine Learning-Based water-level Forecasting and Decision Support System for Ethiopian hydropower plants

A PROJECT REPORT

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Melis Melake

ABSTRACT

Hydropower supplies approximately 90% of Ethiopia's electricity through major plants including the Gibe, GERD, Tekeze, and other hydropower plants. Ethiopian Electric Power's (EEP) Generation, Operation, Monitoring, and Analysis Office requires accurate forecasting of water levels, power generation, and operational loads for grid stability and resource optimization. However, non-linear interactions between hydrological and operational factors, combined with data quality challenges, limit conventional forecasting approaches.

This project presents the development of a Machine Learning–Based Hydropower Forecasting Dashboard designed to support operational and planning activities in hydropower plants for EEP, integrating operational data from fourteen hydropower plants spanning 2013-2025 and meteorological data. A comprehensive data preprocessing pipeline is implemented for all fourteen hydropower plants, involving date standardization, missing value handling, feature scaling using robust normalization techniques, mitigating outliers using the Interquartile Range (IQR) capping method, and feature selection to reduce redundancy and multicollinearity.

Multiple regression-based machine learning models are trained to forecast key target variables, particularly reservoir water level, actual energy production, discharge water, and min/max load of units. Model performance is evaluated using standard statistical metrics such as Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and coefficient of determination (R^2). The trained models generate short- and medium-term forecasts and provide comprehensive visualizations, including actual versus predicted time-series plots, diagnostic boxplots for outlier analysis, and tabular 365-day ahead forecasts for 2026 up to 2030.

These visual outputs are integrated into an interactive Dash-based web application featuring dynamic plant selection, flexible forecast horizons, real-time visualization, and automated anomaly detection flagging deviations exceeding 5%. The system enables users to analyze trends, compare historical and forecasted values, and support operational decision-making. Overall, the system contributes to improved efficiency, reliability, and data-driven planning in Ethiopian hydropower operations.

Keywords: Hydropower Forecasting, Ethiopian Electric Power, Machine Learning, Gradient Boosting, Random Forest, XGBoost, Multi-Output Regression, Anomaly Detection, Interactive Dashboard, Time Series Prediction.

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ACRONYMS AND ABBREVIATIONS

EEP	Ethiopian Electric Power
IQR	Interquartile Range
GB	Gradient Boosting
RF	Random Forest
XGB	Extreme Gradient Boosting
SVR	Support Vector Regression
KNN	K-Nearest Neighbors
DT	Decision Tree
LR	Linear Regression
MSE	Mean Squared Error
RMSE	Root Mean Squared Error
MAE	Mean Absolute Error
R^2	R-Squared
CSV	Comma Separated Values
UI	User Interface
GERD	Grand Ethiopian Renaissance Dam
EDA	Exploratory Data Analysis
ML	Machine Learning
API	Application Programming Interface
SCADA	Supervisory Control and Data Acquisition
NASA	National Aeronautics and Space Administration
GC	Gregorian Calendar
EC	Ethiopian Calendar
KW	Kilowatt
MW	Megawatt
KWH	Kilowatt-hour
m^3/s	Cubic meters per second

1 INTRODUCTION

1.1 Background of the Study

Ethiopia possesses one of the largest hydropower potentials in Africa, with an estimated generation capacity exceeding 45,000 MW, earning the country the designation "the water tower of Africa." Its diverse topography and favorable climatic conditions give rise to fourteen major hydropower plants, including the Abay (Blue Nile), Omo-Gibe, Tekeze, and Awash and others. Hydropower currently accounts for approximately 90% of national electricity generation, forming the backbone of Ethiopia's energy infrastructure and playing a central role in socioeconomic development.

Recognizing this strategic advantage, the Ethiopian government has prioritized hydropower development through successive national growth and transformation frameworks. Significant investments have been made in large-scale projects such as the Gibe (Gibe I, II, and III), the Grand Ethiopian Renaissance Dam (GERD), Tekeze, Finchaa, Koka, Tana Beles, Tis Abay II, Awash II and III, Melka Wakena, Ameri Neshi, and Genelle. These facilities support domestic electrification goals while advancing regional power trade through the Eastern Africa Power Pool.

Despite its vast potential, Ethiopia's hydropower sector faces persistent operational challenges such as seasonal hydrological variability that leads to pronounced fluctuations in reservoir inflows, resulting in reduced generation capacity during dry periods and heightened flood risks during wet seasons. Furthermore, climate change has intensified rainfall uncertainty and altered historical hydrological patterns, undermining the reliability of conventional forecasting and planning approaches. Operational datasets are often fragmented across plants, characterized by inconsistent formats, missing values, and data quality limitations that restrict effective analysis.

The Generation, Operation, Monitoring, and Analysis Office at Ethiopian Electric Power (EEP) requires accurate and timely forecasts of reservoir water levels, energy production, and load dynamics to support optimal reservoir operation, maintenance scheduling, and grid reliability. Traditional statistical and rule-based forecasting methods are insufficient for capturing the complex, nonlinear interactions among meteorological, hydrological, and operational variables inherent in large-scale hydropower systems.

Recent advances in machine learning, particularly ensemble-based techniques such as Gradient Boosting, Random Forest, and XGBoost, provide powerful tools for modeling nonlinear relationships and extracting predictive insights from high-dimensional datasets. Leveraging these methods offers a promising pathway to transform fragmented hydropower data into reliable forecasting, thereby enhancing operational decision-making, strengthening energy security, and supporting Ethiopia's long-term development objectives.

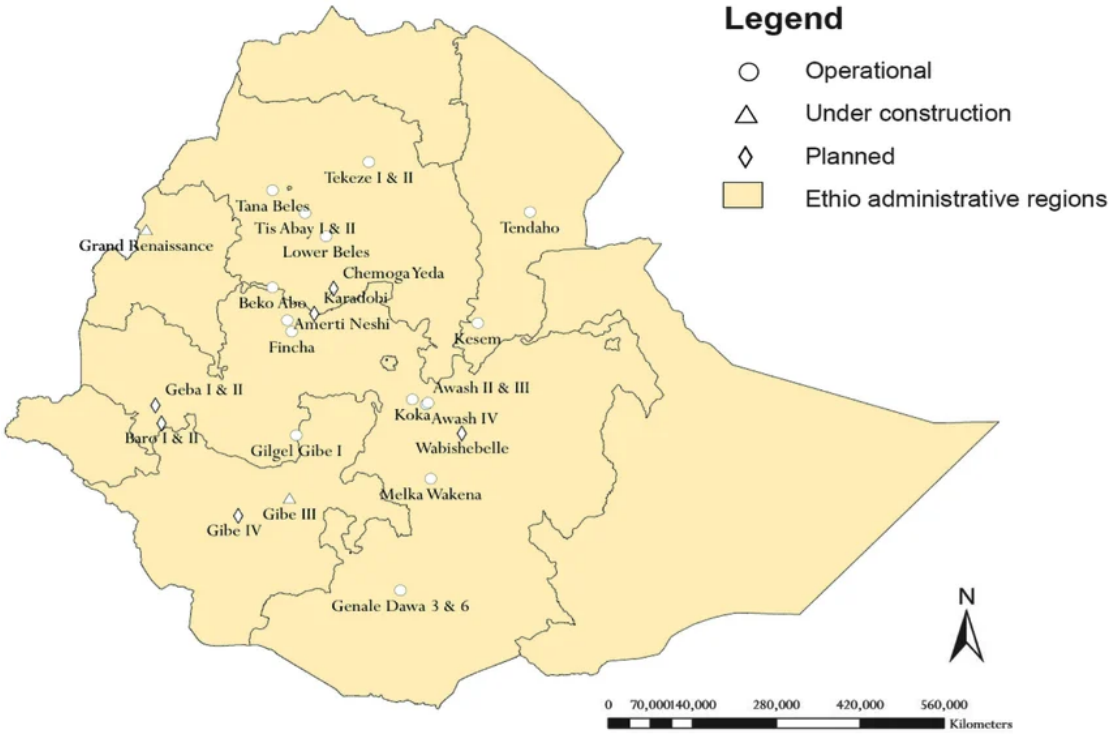


Figure 1.1: Map of major hydropower plants in Ethiopia

1.2 Statement of the Problem

Ethiopian Electric Power's Generation, Operation, Monitoring, and Analysis Office lacks an integrated, data-driven forecasting system capable of accurately predicting key hydropower performance indicators across its major generation facilities. Existing forecasting approaches are largely based on linear assumptions and limited historical trend analysis, which are insufficient for capturing complex and nonlinear interactions among hydrological, meteorological, and operational factors.

The problem is further compounded by persistent data quality challenges, in-

cluding missing values, outliers, and inconsistency in data collection across power plants. These approaches often fail to capture non-linear relationships between hydrological, meteorological, and operational variables, resulting in inaccurate forecasts of energy production, reservoir levels, and auxiliary load requirements. Inaccurate forecasting can lead to inefficient water utilization, increased operational risk, unexpected power shortages, and suboptimal dispatch decisions.

Therefore, the core problem addressed in this project is the lack of an accurate, automated, and integrated forecasting system capable of predicting and forecasting key hydropower operational variables such as energy production, reservoir water level, and discharge using historical plant data and meteorological inputs. Addressing this problem is essential for improving operational efficiency, enhancing planning accuracy, and supporting data-driven decision-making in Ethiopian hydropower plants.

1.3 Objectives of the Study

1.3.1 General Objective

The general objective of this study is to develop an interactive machine learning based reservoir water level forecasting web-based dashboard capable of accurately forecasting key hydropower plants operational indicators across major Ethiopian Electric Power generation facilities, thereby supporting data-driven decision making and optimized resource management within the Generation, Operation, Monitoring, and Analysis Office.

1.3.2 Specific Objectives

The specific objectives of the study are to:

- Integrate relevant datasets and hydropower datasets collected from multiple Ethiopian Electric Power plants
- Design and implement an interactive forecasting dashboard that visualizes historical data, and future forecasts through intuitive time-series plots and tabular outputs
- Identify and model the relationships between hydrological, meteorological, and operational variables affecting hydropower performance using machine learning techniques
- Develop and train forecasting models capable of predicting key target vari-

ables, including actual energy production, reservoir water level, and water discharge

- Implement a multi-step forecasting framework capable of generating medium to long term predictions mainly for water levels, and additionally for energy generation, discharge water from reservoirs and load demand across hydro power plants
- Incorporate anomaly detection mechanisms to automatically identify significant deviations between predicted and observed values and support early operational intervention
- Enhance model transparency and interpretability through feature importance analysis, enabling stakeholders to understand the key drivers of hydropower system performance

1.4 Scope and Limitations of the Study

1.4.1 Scope of the Study

This study focuses on the development and evaluation of a machine learning-based water level forecasting and monitoring system for hydropower operations in Ethiopia. The scope covers fourteen major hydropower plants operated by Ethiopian Electric Power, including facilities within the Gibe (I, II, and III), the Grand Ethiopian Renaissance Dam, and other strategically significant hydro power plants distributed across the country's major river basins.

1.4.1.1 Data and Time Scope

The analysis is based on historical daily operational and meteorological data spanning the period from 2013 to 2025. The study considers critical operational variables such as reservoir water levels, power generation, operational loads, discharge rates, and relevant meteorological factors such as temperature, precipitation, solar radiation, humidity, and wind speed.

1.4.1.2 Target Variables

The forecasting models are developed to predict key hydropower operational variables, including Reservoir water level (main focus in our project), Actual energy production (kWh), Water discharge, Minimum and maximum auxiliary load, unit production (where applicable).

1.4.1.3 Methodological Scope

The study evaluates seven machine learning algorithms and incorporates temporal and lag-based feature engineering to capture seasonal patterns and temporal dependencies.

1.4.1.4 System Scope

The primary delivery of the study is an interactive web-based dashboard that integrates forecasting, visualization, and anomaly detection functionalities to support operational data driven decision-making and extracts useful insight. The system is implemented as an analytical and visualization-based dashboard that presents historical data, forecasting results, and model performance through plots and tables. The dashboard is designed for academic analysis and decision-support demonstration purposes.

1.4.1.5 Geographical and Institutional Scope

The study is designed for Ethiopian Electric Power's (EEP) Generation, Operation, Monitoring, and Analysis Office and the data is conceptually aligned with Ethiopian hydropower plants, methodological datasets and operational conditions.

1.4.2 Limitations of the Study

Despite its contributions, the study is subject to several limitations:

- The quality and availability of historical operational data vary across power plants, with the presence of missing observations and extreme values, measurement errors, or limited historical records requiring imputation and outlier mitigation, which may introduce uncertainty into model predictions.
- The forecasting models trained under the historical patterns remain representative of future conditions, limiting their ability to fully account for abrupt regime shifts caused by climate change or extreme hydrological events.
- The system operates on batch-processed historical data and does not incorporate real-time data streams, which restricts its responsiveness to sudden operational or environmental changes.
- The spatial resolution of the meteorological inputs may not fully capture localized microclimatic variations affecting individual reservoirs.
- Forecast reliability generally decreases as the prediction period extends further into the future.

- Model Generalization: The machine learning models are trained on historical patterns and may not fully capture extreme hydrological events such as droughts or floods that fall outside the training data range.
- The system is designed to preprocess hydropower and meteorological data, train machine learning models, generate forecasts, and visualize results through interactive and static plots.

1.5 Significance of the Study

This study provides significant operational value to Ethiopian Electric Power (EEP) by introducing a data-driven, machine learning-based water level forecasting framework for hydro power plants. Plant operators can better estimate available hydraulic head and expected energy production. The developed system provides EEP's Generation Office with reliable forecasts supporting multiple operational decisions. The anomaly detection system enables early identification of significant deviations for proactive intervention. Improved forecasting accuracy enhances evidence-based planning for reservoir operation, water resource utilization, reduces operational costs and uncertainty, and enhances daily and seasonal planning decisions, turbine scheduling, and load balancing.

From an academic perspective, this study contributes to the growing body of research on data-driven forecasting in hydropower plant systems. It provides a structured workflow that can be used by students and researchers for further studies, comparative analysis of forecasting models, and exploration of advanced machine learning or deep learning techniques.

At the system level, the study strengthens grid management and regional power trade by supporting reliable generation forecasts for domestic load dispatch and export commitments. Beyond operational benefits, the study contributes to organizational capacity building by promoting the adoption of data-driven decision-making practices within Ethiopian Electric Power. Environmentally, improved hydropower forecasting reduces dependence on emergency thermal generation and supports more sustainable water resource management.

2 LITERATURE REVIEW

2.1 Overview of Hydropower Forecasting

Hydropower plant water level forecasting involves predicting key operational variables such as reservoir water levels, power generation, and turbine loads. For Ethiopian Electric Power's Generation, Operation, and Monitoring Office, accurate forecasts are essential for grid stability, water resource management, and energy trading. However, hydropower systems exhibit complex, non-linear behavior due to interactions between rainfall, river flow, evaporation, and operational decisions. This complexity makes forecasting challenging and necessitates advanced modeling approaches beyond traditional statistical methods.

2.2 Traditional Forecasting Methods

Traditional forecasting techniques commonly used in the energy sector include time series models and linear regression. These methods assume that relationships between variables are linear and that data patterns remain stable over time. While they can provide reasonable predictions for short-term periods under normal conditions, they struggle to capture the complex, non-linear behavior of hydropower systems.

Hydropower generation is influenced by many interacting factors including rainfall, river flow, temperature, and operational decisions, which create complicated patterns that simple models cannot adequately represent. Research has shown that traditional linear approaches perform poorly when applied to hydrological and energy data that exhibit strong seasonal variations and sensitivity to climate variability.

At Ethiopian Electric Power, current forecasting practices rely primarily on historical trend analysis and rule-based methods developed from operator experience. These approaches, while valuable, are insufficient for capturing the complex dynamics observed across Ethiopia's diverse hydropower plants, leading to suboptimal operational decisions and reduced forecast accuracy.

2.3 Machine Learning in Time Series Prediction

Machine learning algorithms have gained prominence for time series forecasting due to their ability to model non-linear relationships without requiring predefined assumptions about data distributions. Decision Trees and K-Nearest Neighbors provide interpretable baselines but often suffer from high variance or poor scalability. Support Vector Regression (SVR) can model complex boundaries but is computationally intensive for large datasets. Research demonstrates that machine learning approaches consistently outperform traditional statistical methods for renewable energy forecasting tasks, making them suitable for implementation at EEP's Generation Office.

2.4 Ensemble Methods

Ensemble methods combine multiple learners to improve prediction accuracy and robustness.

2.4.1 Random Forest

Random Forest, introduced by Breiman (2001), builds numerous decision trees on bootstrap samples and averages their predictions. It reduces overfitting, handles high dimensional data, and provides reliable feature importance estimates. This method is particularly valuable for identifying which operational variables most influence hydropower generation at EEP plants.

2.4.2 Gradient Boosting

Gradient Boosting, developed by Friedman (2001), constructs trees sequentially where each new tree corrects errors from previous ones. This iterative approach achieves high accuracy and has been successfully applied in similar energy forecasting contexts.

2.4.3 XGBoost

XGBoost (Extreme Gradient Boosting), proposed by Chen and Guestrin (2016), is an optimized implementation of gradient boosting with regularization, parallel processing, and handling of missing values. It offers an excellent balance of speed and accuracy, making it suitable for operational forecasting at EEP where timely predictions are needed for daily decision-making.

2.5 Related Works on Hydropower Forecasting

Several studies have applied machine learning techniques to hydropower forecasting, providing valuable insights for this project.

- **Dires, Amelin & Bekele (2023) Inflow Scenario Generation for the Ethiopian Hydropower System:** Synthetic historical inflow time series for major Ethiopian reservoirs over a ten-year period (2010–2020) using satellite precipitation data and hydropower generation records by using SARIMA models and Monte Carlo simulation to generate reservoir water level and stochastic inflow scenarios at daily and weekly time resolutions.
- **Youssef Faouzi (2019) – Hydropower Forecast in Ethiopia Using the HBV Model (master's thesis):** This research applies the rainfall-runoff hydrological model (HBV) to forecast hydropower generation in the Omo-Gibe basin of Ethiopia by simulating reservoir inflow and water balance.
- **Dires (2023) – Long-Term Hydropower Planning for Ethiopia with Inflow Uncertainty:** A paper comparing stochastic and deterministic models for Ethiopian hydropower planning using inflow forecasting and rolling horizon frameworks.

2.6 Research Gap

Ethiopian Electric Power lacks an integrated, plant-specific forecasting system that:

- Addresses the significant data quality challenges (missing values, outliers, inconsistent formats) prevalent across its 14 major hydropower plants
- Provides multi-output predictions for correlated variables (water level, power generation, maximum and minimum load) simultaneously, maintaining physical consistency between forecasts
- Delivers operational forecasts through an accessible, interactive interface for the Generation Office staff
- Incorporates automated anomaly detection to alert operators when significant deviations between predicted and actual values occur

3 METHODOLOGY

3.1 Data Collection and Description

3.1.1 Power Plants Dataset

This study utilizes historical operational data collected for fourteen major hydropower plants operated by Ethiopian Electric Power from Ethiopian Electric Power's (EEP) Generation, Operation, Monitoring, and Analysis Office. The dataset comprises daily operational records spanning the period from 2013 to 2025, although data availability varies across individual plants due to differences in commissioning dates, instrumentation upgrades, and data recording practices. The primary operational variables include water level, unit power outputs (U1 pr, U2 pr, U3 pr), maximum and minimum active load, actual energy generation, and discharge rate.

3.1.2 Meteorological Data

The meteorological data obtained from NASA POWER include temperature (T2M), precipitation (PRECTOTCORR), relative humidity (RH2M), surface pressure (PS), wind speed (WS10M), and solar radiation (ALLSKY SFC SW DWN) spanning the period from 2013 to 2025.

3.2 Data Preprocessing

We design a general preprocessing pipeline for all fourteen Ethiopian hydropower plants to have a standard, repeatable workflow that can be applied to any power plant with minimal adjustments. We illustrate it using Gibe 1 as an example.

3.2.1 Data Overview

The Gibe 1 reservoir dataset contains 4,383 daily records spanning approximately 12 years from 2013 to 2025. It includes 15 columns capturing operational variables including three units (U1 pr, U2 pr, U3 pr), maximum and minimum auxiliary loads, water level of the reservoir, actual energy produced, reservoir discharge, Gregorian and Ethiopian calendar dates, and meteorological features.

3.2.2 Duplicate Record Handling

To ensure data integrity, the dataset was checked for duplicate records that could bias model training or distort statistical analysis. The analysis revealed no duplicate entries in the primary dataset, confirming that the data collection process for Gibe 1 was consistent and reliable.

3.2.3 Date-Time Conversion (Date Standardization)

Raw datasets contained inconsistent date representations, including Gregorian calendar dates and Ethiopian calendar notations. For modeling purposes, we created a numeric feature (Date GC float) from the Gregorian calendar date representing days since the first observation.

3.2.4 Conversion of Non-Numeric Columns

Some columns in the Gibe 1 dataset, such as ACTUAL ENERGY(KWH), Date GC, Date EC, and Discharge, were stored as object types due to formatting issues. All non-numeric columns were converted to numeric values.

3.2.5 Column Name and String Cleaning

To ensure consistency and prevent errors, column names and object-type values were cleaned by removing non-word characters, replacing spaces with underscores, and stripping leading/trailing spaces.

3.2.6 Feature Engineering

A new numeric time feature (Date GC float) was created to make the data suitable and readable for ML models while preserving the sequential indexing of dates.

3.2.7 Dropping Original Date Columns

After creating the numeric time feature, the original date columns (Date GC, Date EC) were dropped as they were no longer needed for modeling.

3.2.8 Missing Value Treatment

For each relevant variable, the percentage of missing observations was calculated and handled according to predefined thresholds. Variables with less than 40% missing values were imputed using the mean. Variables with 40% to 60% missing

values were filled with zero. Variables with more than 60% missing values were removed.

Table 3.1: Data Type Conversion Results - Before and After

Column Name	**Before Conversion**	**After Conversion**
Date GC	datetime64[ns]	datetime64[ns]
Date EC	object	int64
U1 pr	int64	int64
U2 pr	float64	float64
U3 pr	float64	float64
Max ALoad	float64	float64
Min ALoad	float64	float64
Water Level	float64	float64
T2M	float64	float64
PRECTOTCORR	float64	float64
ACTUAL ENERGY(KWH)	object	—
ACTUAL ENERGY KWH	—	float64
Discharge	object	float64
ALLSKY SFC SW DWN	float64	float64
RH2M	float64	float64
WS2M	float64	float64
Date GC float	float64	float64

Table 3.2: Missing Value Statistics Before Treatment

Column Name	NonNullCount	MissingCount	MissingPercent	Skewn
Date GC	4383	0	0.00	0.0
Date EC	4383	0	0.00	0.0
U1 pr	4383	0	0.00	-0.9
U2 pr	4382	1	0.02	-0.4
U3 pr	4382	1	0.02	-0.8
Max ALoad	4018	365	8.33	-0.3
Min ALoad	4018	365	8.33	1.0
Water Level	4383	0	0.00	-0.2
T2M	4383	0	0.00	0.5
PRECTOTCORR	4383	0	0.00	2.7
ACTUAL ENERGY(KWH)	4383	0	0.00	0.0
Discharge	4383	0	0.00	0.0
ALLSKY SFC SW DWN	4383	0	0.00	-0.7
RH2M	4383	0	0.00	-0.8
WS2M	4383	0	0.00	1.0

3.2.9 Outlier Detection and Treatment

3.2.9.1 IQR-Based Outlier Detection

Outliers were identified using the Interquartile Range (IQR) method. For each numeric variable, observations falling below Q1 − 1.5 × IQR or above Q3 + 1.5 × IQR were classified as outliers.

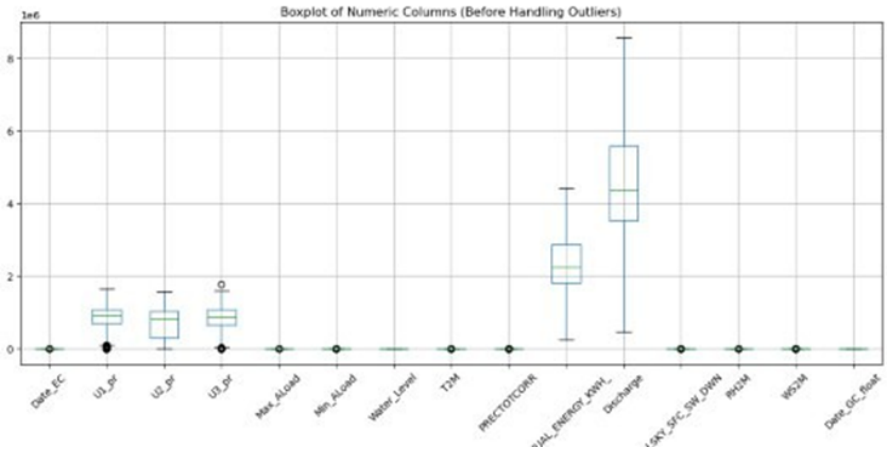


Figure 3.1: Boxplot of numeric columns before handling outliers

3.2.10 Outlier Capping (Winsorization) Results

Rather than removing outliers, outlier treatment was performed using capping (winsorization). Values exceeding IQR-based thresholds were replaced with the corresponding lower or upper bounds.

Table 3.3: Outlier Capping Results for Gibe I Dataset

Variable	**Original Outliers**	**After Minimization**
U1 pr	232	0
U2 pr	0	0
U3 pr	302	0
Max ALoad	9	0
Min ALoad	51	0
Water Level	0	0
T2M	33	0
PRECTOTCORR	237	0
ACTUAL ENERGY KWH	0	0
Discharge	0	0
ALLSKY SFC SW DWN	69	0
RH2M	26	0
WS2M	129	0
Date GC float	0	0

3.2.11 Correlation Analysis

The correlation heatmap illustrates the linear relationships among operational, hydrological, and meteorological variables prior to model development.

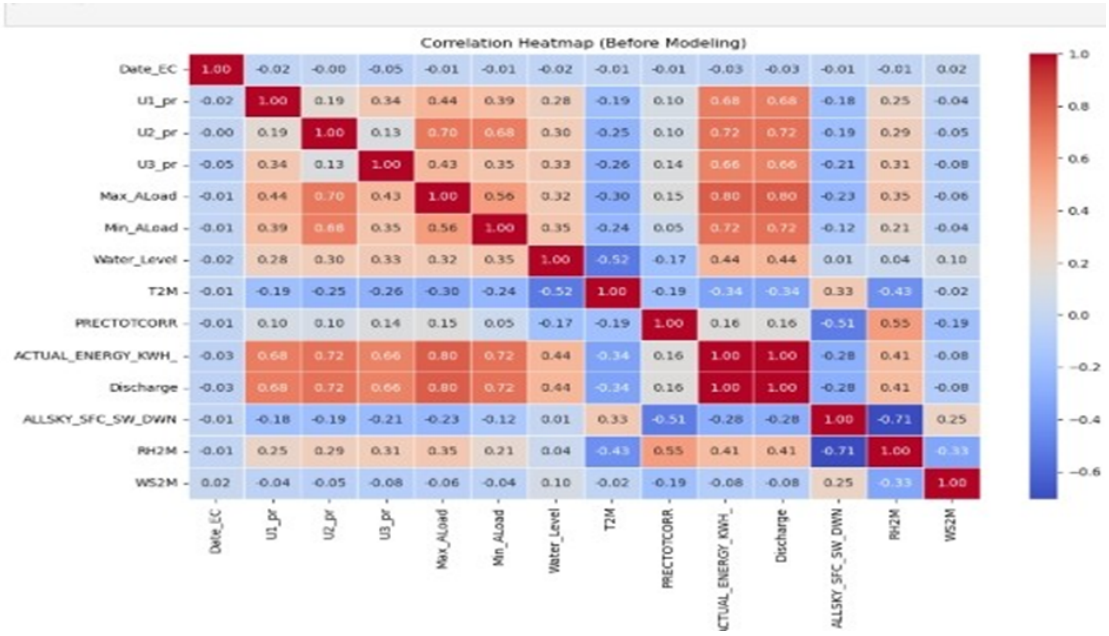


Figure 3.2: Correlation Heatmap of Operational, Hydrological, and Meteorological Variables

3.2.12 Lag Features

Lag features were created at 1, 2, 3, 7, 14, and 30-day intervals for all target variables to capture temporal dependencies. Analysis confirmed that **lag features dominate predictive power**, contributing approximately 45% of total feature importance in the final models.

3.2.13 Target Variable Selection

Target variables were identified using hydropower-specific keywords including 'level', 'U', 'energy', 'discharge', 'auxiliary', and 'load'. For the Gibe I dataset, eight target variables were selected: U1 pr, U2 pr, U3 pr, Max Load, Min Load, Water Level, Actual Energy KWH, and Discharge.

3.2.14 Train-Test Split

A train-test split was adopted with the earliest 80% of observations (3,214 samples) used for training and the most recent 20% (804 samples) reserved for testing.

3.3 Machine Learning Models

3.3.1 Model Configurations

The following machine learning models were implemented and evaluated:

- **Decision Tree Regressor:** Tree-based model that splits data based on feature thresholds
- **Random Forest Regressor:** Ensemble of decision trees that reduces variance
- **Linear Regression:** Standard linear model for baseline comparison
- **Support Vector Regressor (SVR):** Uses kernel functions to model nonlinear relationships
- **Gradient Boosting Regressor:** Builds trees sequentially to minimize prediction errors
- **XGBoost Regressor:** Optimized gradient boosting for speed and performance
- **K-Nearest Neighbors (KNN):** Predicts based on nearest neighbors in feature space

3.3.2 Multi-Output Regression Framework

A multi-output regression framework was employed to enable simultaneous prediction of all target variables, maintaining physical consistency between correlated operational quantities.

3.4 Model Evaluation Metrics

3.4.1 R^2 (Coefficient of Determination)

$$R^2 = 1 - \frac{\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}{\sum_{i=1}^{n}(y_i - \bar{y})^2}$$

3.4.2 Mean Squared Error (MSE)

$$MSE = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2$$

3.4.3 Root Mean Squared Error (RMSE)

$$RMSE = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}$$

3.4.4 Mean Absolute Error (MAE)

$$MAE = \frac{1}{n}\sum_{i=1}^{n}|y_i - \hat{y}_i|$$

3.4.5 Model Performance Comparison

Table 3.4: Model Performance Comparison for Gibe I Dataset

Model	**R^2**	**MSE**	**RMSE**	**MAE**
Gradient Boosting	0.8488	0.0080	0.0896	0.0550
Random Forest	0.8333	0.0091	0.0953	0.0556
XGBoost	0.8238	0.0096	0.0979	0.0605
Decision Tree	0.7403	0.0141	0.1187	0.0648
KNN	0.6570	0.0197	0.1402	0.0948
SVR	0.6423	0.0204	0.1430	0.0941
Linear Regression	0.4846	0.0310	0.1759	0.1324

4 RESULTS AND DISCUSSION

From this study, an interactive forecasting dashboard system was successfully developed. The system allows users to select a specific hydropower plant, target variable, and forecasting time period. Based on these selections, the dashboard generates forecasts and presents them alongside actual historical values through clear graphical visualizations.

4.1 Discussion of Findings

4.1.1 Implications of Feature Importance

Feature importance analysis provides critical insight into the factors that most strongly influence water level forecasting at the Gibe 1 Hydropower Reservoir. Using the random forest regression model, feature importance scores were derived to quantify the contribution of each predictor variable.

Among the most influential features, historical water level emerged as the primary predictor. The high importance of lag-related and time-derived features indicates that the model effectively captures seasonal and cyclical patterns inherent in the Gibe 1 reservoir system.

Meteorological variables, particularly precipitation (PRECTOTCORR), temperature (T2M), relative humidity (RH2M), wind speed (WS2M), and solar radiation, also contributed significantly to model performance.

4.2 Dashboard Implementation Results

4.2.1 User Interface

The dashboard features a clean, intuitive interface with header containing title and notification icon, controls panel for user input, main visualization area, action buttons for data export, and an expandable Action Center.

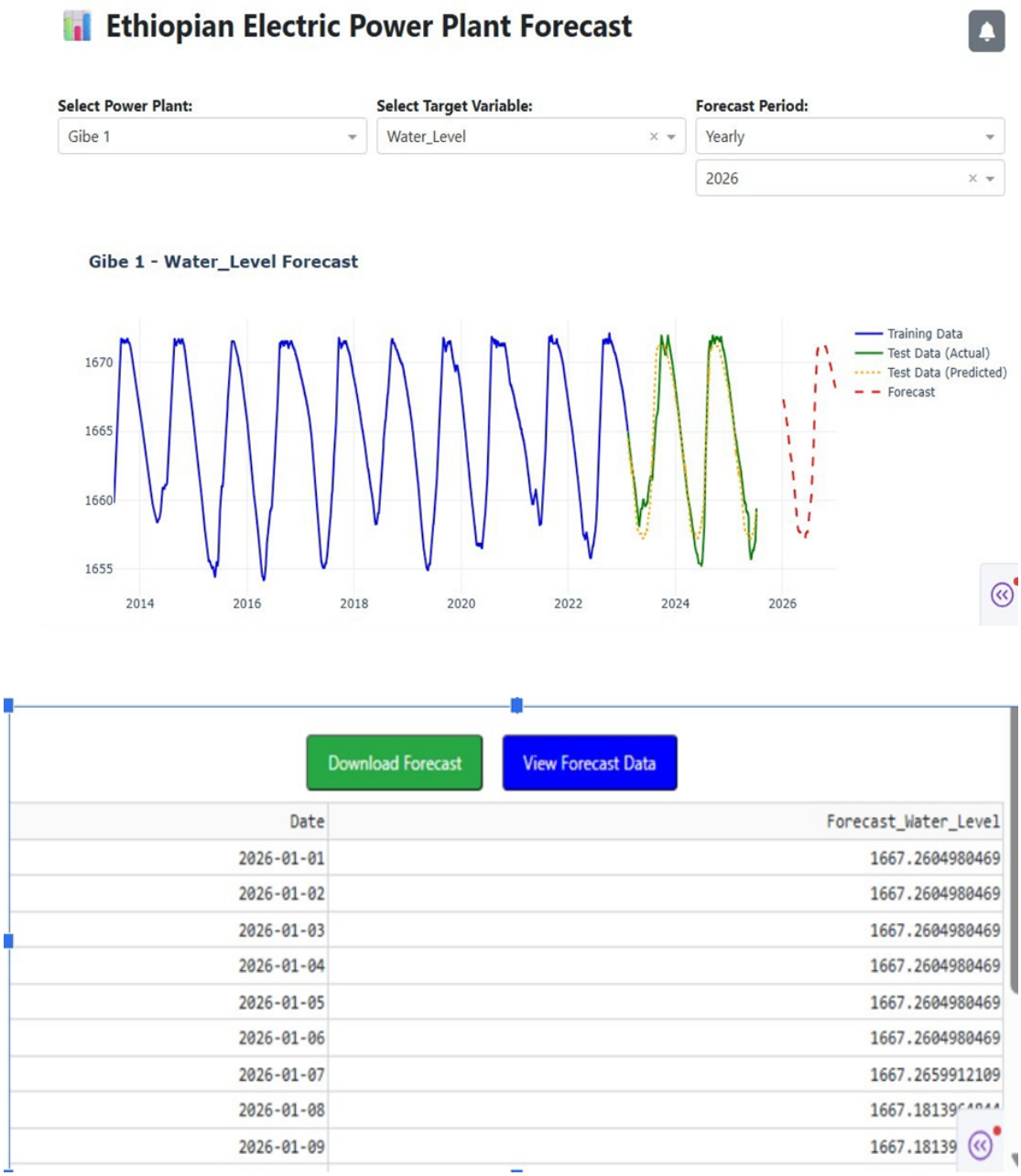


Figure 4.1: Ethiopian Electric Power Plant Forecast Dashboard - Gibe 1 Water Level Prediction

4.2.2 Interactive Controls

Interactive controls include power plant selector with 14 options, dynamically populated target variable selector, forecast period selector (daily, weekly, monthly, yearly), view forecast button, and download button.

4.2.3 Visualization Features

The main visualization displays training data (blue line), test data actual (green line), test data predicted (orange dashed line), and future forecast (red dashed line) with interactive tooltips and zoom capabilities.

Figure 4.1 presents the water level forecasting results for the Gibe I Hydropower Plant, showing the comparison between historical training data, actual test observations, model predictions, and the future forecast period. The visualization clearly illustrates the model's ability to learn seasonal patterns, capture short-term fluctuations, and generate realistic future water level projections. The training data (blue line) exhibits a strong and consistent seasonal cycle, characterized by periodic peaks and troughs corresponding to wet and dry seasons. This repeating pattern indicates stable hydrological behavior of the Gibe I reservoir over time. The smooth continuity of the training curve confirms that the preprocessing steps successfully removed noise and ensured temporal consistency in the dataset.

During the test period, the actual observed water levels (green line) closely follow the historical seasonal pattern. The predicted values (orange dotted line) align well with the actual test data, particularly in capturing the timing of rising and falling water levels. The model accurately tracks both peak levels and minimum reservoir depths, demonstrating strong generalization capability and low prediction error during the validation phase. Minor deviations between predicted and actual values are observed near extreme peaks and troughs, which is expected due to sudden hydrological variations and operational water releases that are difficult to fully capture in data-driven models.

The forecasted water levels for the future period (red dashed line) maintain the established seasonal trend, with projected peaks and declines occurring at consistent intervals. The forecast remains within a realistic range observed in historical records, indicating that the model avoids unrealistic over- or under-estimation. This stability suggests that the forecasting system is suitable for medium-term operational planning.

Overall, the results demonstrate that the developed machine learning model effectively captures the temporal dynamics of the Gibe I reservoir. The close agreement between actual and predicted values confirms the reliability of the model for water level forecasting. .

4.2.4 Anomaly Detection

An alert and notification system highlights periods of high deviation (5%) between forecasted and historical actual values. Alerts are presented through a notification icon badge with count indicator.

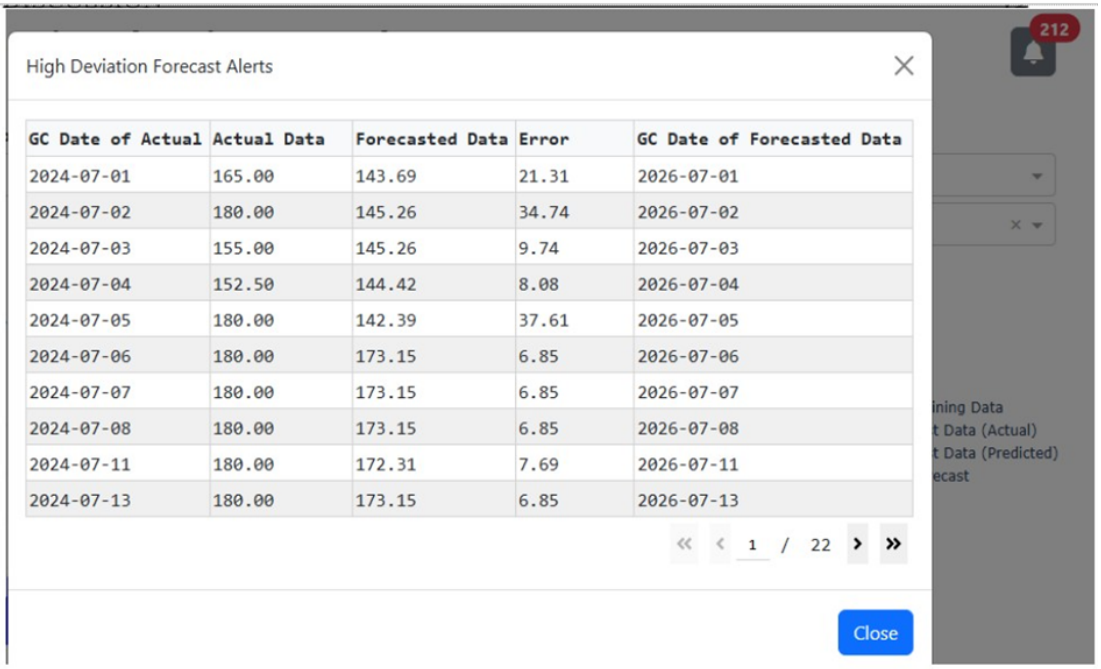


Fig 4.1 The forecasted data for gibe 1 water level in 2026

Figure 4.2: High Deviation Forecast Alerts for Gibe 1 with errors exceeding 5%

4.2.5 2025-2030 Forecast Results

The recursive forecasting framework successfully generated 365-day ahead predictions for 2025 to 2030 across all target variables. Forecasts show expected seasonal patterns with higher water levels during wet seasons (Kiremt).

4.3 Challenges and Limitations of the Project

Despite successful development, the project encountered several technical and operational challenges. Tana Beles II water level and other target variables were not forecasted accurately. Unlike Gibe 1, which showed stable prediction performance, Tana Beles II did not perform as well. Although the system was designed to forecast multiple operational variables, forecasting was successfully implemented only for reservoir water level; other targets were excluded due to unreliable predictions.

5 CONCLUSION AND RECOMMENDATIONS

5.1 Conclusion

This project successfully developed and deployed a machine learning-based forecasting dashboard for the Ethiopian Electric Power (EEP) Generation, Operation, Monitoring, and Analysis Office. The system integrates historical and operational data from 14 major hydropower plants and achieves up to 84.8% prediction accuracy using Gradient Boosting. The preprocessing pipeline effectively handled missing values and capped outliers across key operational metrics, creating a reliable analytical foundation previously unavailable to EEP.

Feature importance analysis confirmed that lag variables (45%) and seasonal patterns (25%) are the most influential predictors, validating the applied feature engineering approach. The interactive dashboard now enables operators to make data-driven decisions regarding turbine scheduling, water release timing, maintenance planning, and grid management.

The study demonstrates that ensemble machine learning methods, particularly Gradient Boosting and Random Forest, are crucial for capturing the complex, nonlinear dynamics of hydropower systems. By transforming fragmented historical data into actionable forecasting intelligence, this system provides EEP with a scalable foundation for advanced analytics, supporting reliable electricity generation, improved water resource management, and Ethiopia's regional energy export ambitions. Overall, the work addresses critical operational challenges and establishes a platform for informed decision-making across the country's major hydropower facilities.

5.2 Recommendations (Future Work)

- Train operators and managers on dashboard usage, forecast interpretation, and proper response to anomaly alerts
- Integrate dashboard with existing SCADA systems for real-time sensor data enabling automatically updating forecasts
- Implement ensemble forecasting combining predictions from multiple models for more reliable forecasts with confidence indicators
- Incorporate long-term climate predictions like ENSO to improve seasonal forecasting accuracy
- Implement consistent data recording protocols across all plants with unified column names, units, and date formats
- Develop REST APIs for seamless data exchange between the dashboard and other EEP systems
- Explore deep learning approaches including LSTM and GRU networks for time series prediction
- Make all hydropower plants and their respective target variables forecasted accurately in one dashboard system
- Integrate other relevant datasets like satellite imagery, meteorological, and geographical data
- Develop mobile-optimized version for field operations

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APPENDIX

Appendix A: DASHBOARD USER GUIDE

A.1 Getting Started

1. Select a power plant from the dropdown menu
2. Choose target variable for forecasting
3. Select forecast period (Daily/Weekly/Monthly/Yearly)
4. View forecast graph with training, test, and prediction data

A.2 Anomaly Detection

The notification icon displays the count of high-deviation alerts (errors exceeding 5%). Click to view detailed anomaly table with actual values, predicted values, and error percentages.

Appendix B: DATA DICTIONARY

Table 1: Data Dictionary for Gibe I Dataset

Column Name	**Description**	**Unit**	**Data Type**
Date GC	Gregorian calendar date	Date	datetime
Date EC	Ethiopian calendar date	Date	int
U1 pr	Unit 1 power generation	KW	float
U2 pr	Unit 2 power generation	KW	float
U3 pr	Unit 3 power generation	KW	float
Max ALoad	Maximum active load	MW	float
Min ALoad	Minimum active load	MW	float
Water Level	Reservoir water level	meters	float
T2M	Temperature at 2 meters	°C	float
PRECTOTCORR	Corrected precipitation	mm/day	float
ACTUAL ENERGY KWH	Actual energy generated	KWH	float
Discharge	Water discharge rate	m^3/s	float
ALLSKY SFC SW DWN	Solar radiation	$kWh/m^2/day$	float
RH2M	Relative humidity	%	float
WS2M	Wind speed at 2 meters	m/s	float
Date GC float	Days since first date	days	float

Appendix C: DESCRIPTIVE STATISTICS FOR GIBE I DATASET

Variable	Count	Mean	Std	Min	25%	50%	75%	Max
U1_pr	4383	852,237	320,089	0	686,260	909,420	1,079,700	1,644,340
U2_pr	4382	682,770	446,136	0	299,995	805,780	1,030,180	1,561,700
U3_pr	4382	820,946	339,482	0	655,360	867,200	1,071,815	1,770,700
Max_ALoad	4018	141.4	35.7	14.3	120.0	135.0	180.0	320.0
Min_ALoad	4018	47.9	21.3	0.0	30.0	45.0	60.0	150.0
Water_Level	4383	1664.8	5.4	1654.1	1659.9	1665.3	1670.1	1672.2
T2M	4383	18.5	1.8	13.3	17.2	18.1	19.7	24.4
PRECTOTCORR	4383	5.3	7.5	0.0	0.2	2.4	7.7	74.5

Figure 1: Descriptive Statistics for Key Variables - Gibe I Dataset